

Core CS — OOP, DBMS, OS & Networks Notes

PlacementPrep Study Notes

1. Object-Oriented Programming (OOP)

Four pillars every interviewer expects you to explain with examples.

- Encapsulation: bundling data and methods, restricting direct access to internal state.
- Abstraction: exposing only essential features, hiding implementation detail.
- Inheritance: a class acquiring properties/behavior of another class.
- Polymorphism: same interface, different underlying behavior (overloading/overriding).

2. DBMS — Normalization

Normalization reduces data redundancy and improves integrity.

- 1NF: atomic column values, no repeating groups.
- 2NF: 1NF + no partial dependency on a composite key.
- 3NF: 2NF + no transitive dependency on non-key attributes.
- BCNF: stricter version of 3NF for certain edge cases with multiple candidate keys.

3. Operating Systems

Know process scheduling, memory management, and deadlocks.

- FIFO page replacement can suffer from Belady's Anomaly (more frames, more page faults).
- Deadlock requires 4 conditions: mutual exclusion, hold and wait, no preemption, circular wait.
- Breaking any one of the four conditions can prevent a deadlock.

4. Computer Networks

Focus on the OSI/TCP-IP layers and common protocols.

- TCP/IP model layers: Application, Transport, Internet, Link.
- The Internet layer handles logical addressing and routing (IP).
- TCP is connection-oriented and reliable; UDP is connectionless and faster but unreliable.

Practice Tip

- Draw diagrams (ER diagrams, OSI layers) on paper/whiteboard when explaining — it shows clarity.
- Relate concepts to a project you've built, e.g., 'I used normalization in my DB design for...'